

PORTFOLIO

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Section 1: Demonstration of Core Knowledge in the Field

This section demonstrates my core knowledge in electrical and computer engineering through four skill areas: CMOS device/process simulation, phase-locked loop design, sensor systems, and communication systems. Each area includes evidence from simulations, figures, or project results rather than only a general concept description.

1.1 CMOS Devices, Process Simulation, and Inverter Operation

A CMOS inverter is built from complementary PMOS and NMOS transistors. When the input is low, the PMOS device pulls the output high. When the input is high, the NMOS device pulls the output low. This complementary action creates rail-to-rail output swing and low static power because the direct current path from VDD to ground is ideally off in the stable logic states. The important engineering skill is not only knowing this concept, but also understanding how device fabrication, threshold voltage, oxide thickness, and doping affect the inverter behavior.

CMOS Inverter

Inverter operation

- When $V_{in} = V_{dd}$, $V_{out} = 0V$
NMOS = "ON" and PMOS = "OFF"
- When $V_{in} = 0$, $V_{out} = V_{dd}$
NMOS = "OFF" and PMOS = "ON" ($V_{SG} < -0.6V$)
- Very little current flows in either output state. (Only during switching)
- Output swings all the way from rail to rail
- Power dissipation is only a function of switching frequency: $P = f C_L V_{dd}^2$
- Noise margins better because transition region is steeper than for NMOS

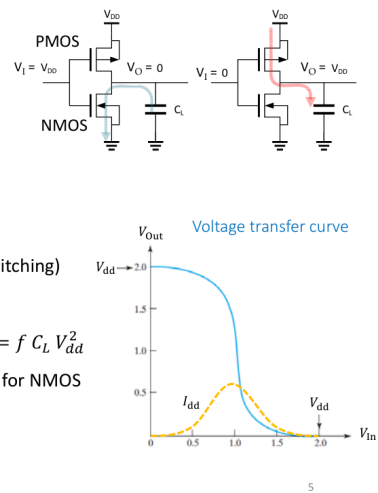


Figure 1. CMOS inverter operation and voltage transfer curve used to connect PMOS/NMOS switching to circuit behavior. Adapted from ECE 612 CMOS lecture notes.

In ECE 612, I simulated the CMOS process flow using Silvaco ATHENA/ATLAS. First, I examined LOCOS isolation. The field oxide and bird's-beak shape show how wet oxidation laterally grows under the nitride mask. This matters because isolation prevents leakage paths between neighboring transistors. The simulation gave me a physical connection between process integration and the electrical behavior of a CMOS circuit.

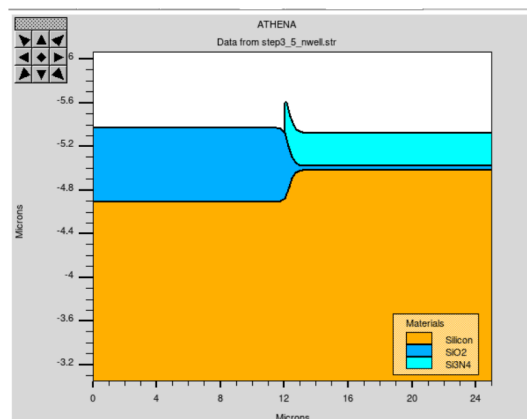


Figure 2. LOCOS simulation result showing field oxide and bird's-beak formation from my ECE 612 CMOS inverter process simulation.

After completing the process flow, I inspected the final structure with regions, junctions, and doping contours. This verified that the p-well, n-well, polysilicon gate, source/drain implants, field oxide, and metal contacts were formed in the expected locations. The doping contour plot helped me see why NMOS and PMOS threshold voltages differ and why well engineering is required for proper CMOS operation.

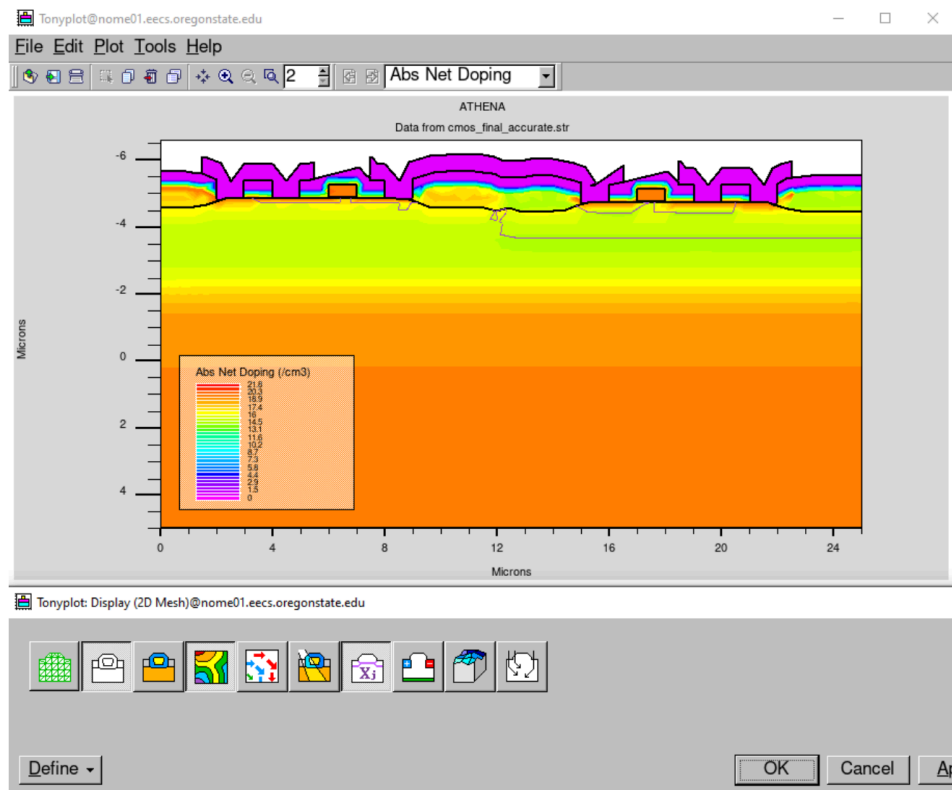


Figure 3. Final simulated CMOS structure with material regions and doping contours, showing the completed twin-well CMOS inverter process.

I then used ATLAS to simulate the NMOS and PMOS transfer characteristics. The NMOS current increased sharply after the positive threshold voltage, while the PMOS current increased for negative gate bias. This verified that both transistors formed working channels. I also extracted threshold voltage using a drain current criterion of 1×10^{-7} A. The extracted results were approximately $V_{th,NMOS} = +1.55$ V and $V_{th,PMOS} = -0.53$ V. These values confirmed correct transistor polarity and demonstrated that channel doping, oxide thickness, and well structure strongly affect switching behavior.

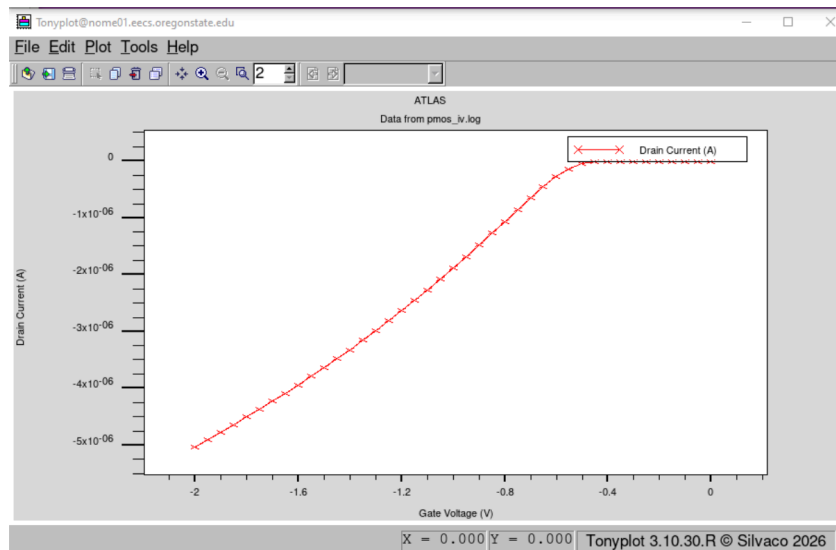
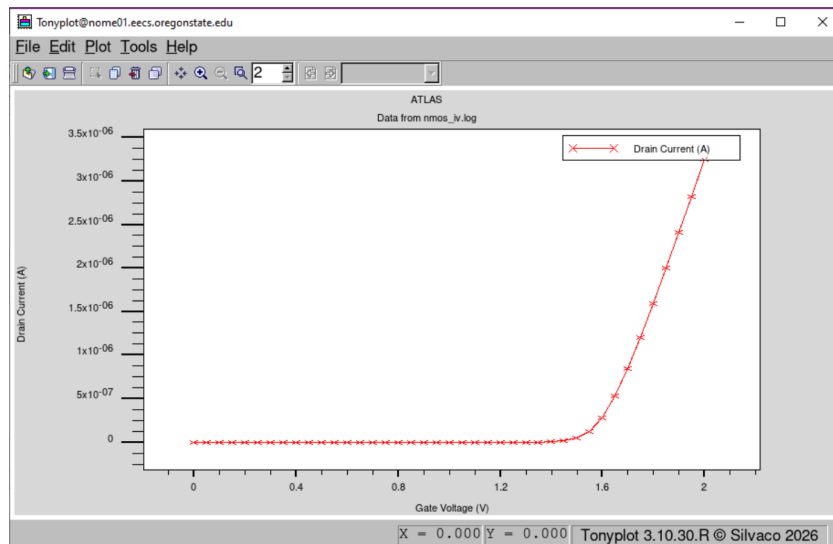


Figure 4. Simulated NMOS and PMOS transfer characteristics from ATLAS.

```

Vth_NMOS=1.53546343
EXTRACT>
EXTRACT> #PMOS Vth
EXTRACT> quit
EXTRACT> init inf="pmos_iv.log"
EXTRACT> extract name="Vth_PMOS" x.val from curve(v."gate", abs(i."drain")) where y.val=1.0e-7
datafile="task5_vth.dat"
Vth_PMOS=-0.531241357
EXTRACT>
EXTRACT> # =====
EXTRACT> # Transfer Characteristics
EXTRACT> # =====
tonyplot nmos_iv.log
tonyplot pmos_iv.log
EXTRACT>
EXTRACT> quit

```

Figure 5. Threshold-voltage extraction results from the CMOS process simulation.

Finally, I connected the device-level knowledge to circuit-level behavior by using a DC sweep of the inverter voltage transfer characteristic. The switching point occurs near the center of the transition region, where both transistors conduct briefly and the gain is highest. This analysis showed me how transistor sizing and threshold voltage control noise margin, switching symmetry, and static power.

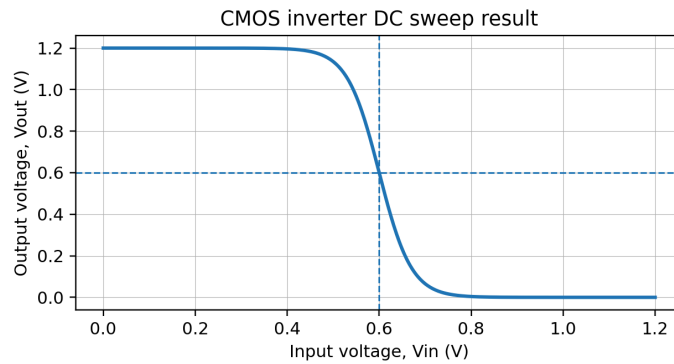


Figure 6. CMOS inverter voltage transfer curve used to identify the switching region and noise-margin behavior.

1.2 Phase-Locked Loop Design and High-Speed Clock Generation

A phase-locked loop is a feedback system that aligns an output clock with a reference clock. I studied a wide-tuning PLL architecture for a 6-12 GHz target range. The system uses the standard chain of phase-frequency detector, charge pump, loop filter, voltage-controlled oscillator, and feedback divider. The main design challenge was achieving a wide tuning range and low jitter while using a ring oscillator, which is highly tunable but has high phase noise.

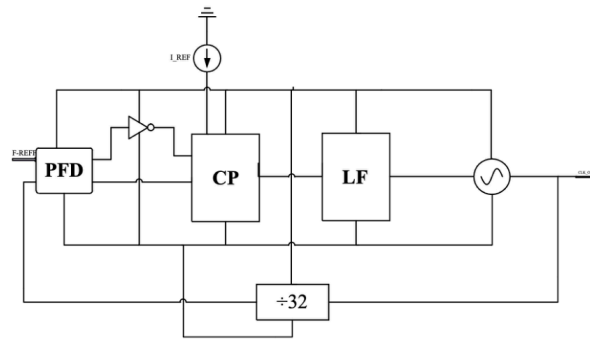


Figure 7. PLL top-level architecture showing PFD, charge pump, loop filter, VCO, and divide-by-32 feedback path.

The VCO was designed as a current-starved ring oscillator. The tuning data showed that the output frequency reached approximately 6 GHz at $V_{ctrl} = 400$ mV and 12 GHz at $V_{ctrl} = 480$ mV. The corresponding K_{VCO} values were about 65 GHz/V and 76 GHz/V. This high K_{VCO} is helpful for wide tuning, but it also makes the control node sensitive to noise. Therefore, loop-filter design and phase-noise analysis became critical parts of the PLL design.

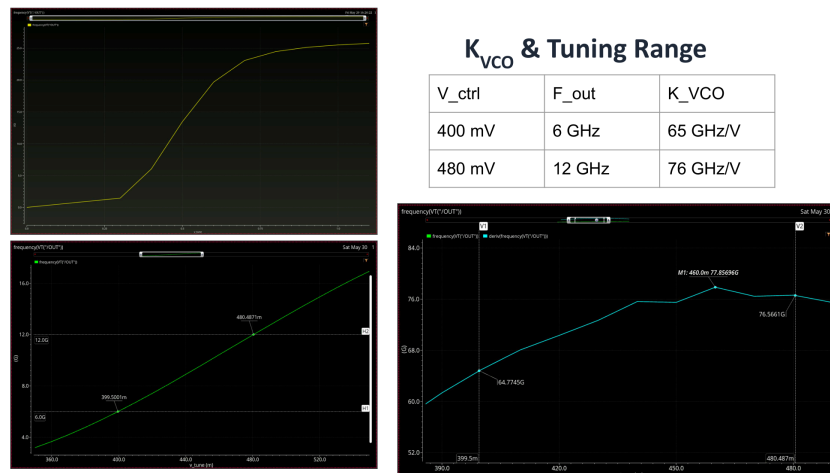


Figure 8. VCO tuning range and K_{VCO} data showing operation from 6 GHz to 12 GHz.

For the loop filter, the design used $R = 65$ ohms, $C1 = 1$ nF, and $C2 = 2$ pF. The charge pump was intentionally scaled to provide a large current so that the loop bandwidth could suppress ring-VCO noise while keeping resistor thermal noise lower. This was a design tradeoff: power was increased to improve phase-noise and jitter performance.

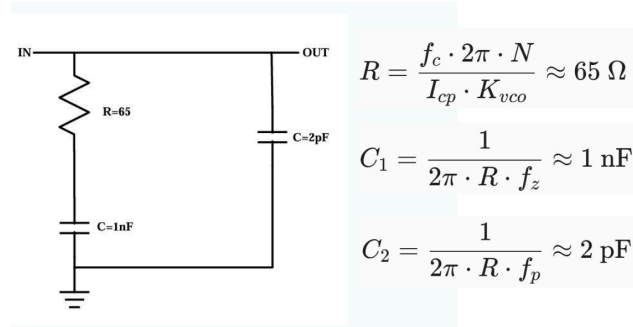


Figure 9. PLL loop-filter design with $R = 65$ ohms, $C1 = 1$ nF, and $C2 = 2$ pF.

MATLAB-style loop-gain verification was used to check stability. At the 12 GHz corner, the loop bandwidth was about 49.5 MHz with phase margin about 84.9 degrees. At the 6 GHz corner, the bandwidth was about 41.9 MHz with phase margin about 84.4 degrees. These results show that the PLL had a stable phase-margin plateau across the tuning range.

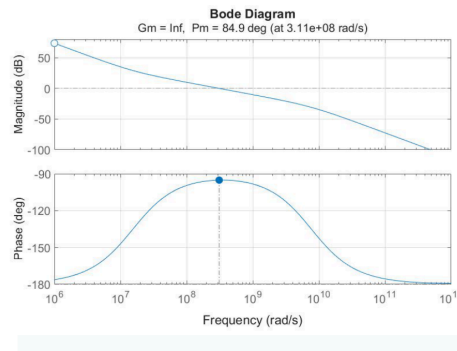


Figure 10. Bode plot verification at the 12 GHz corner, showing approximately 49.5 MHz bandwidth and 84.9 degree phase margin.

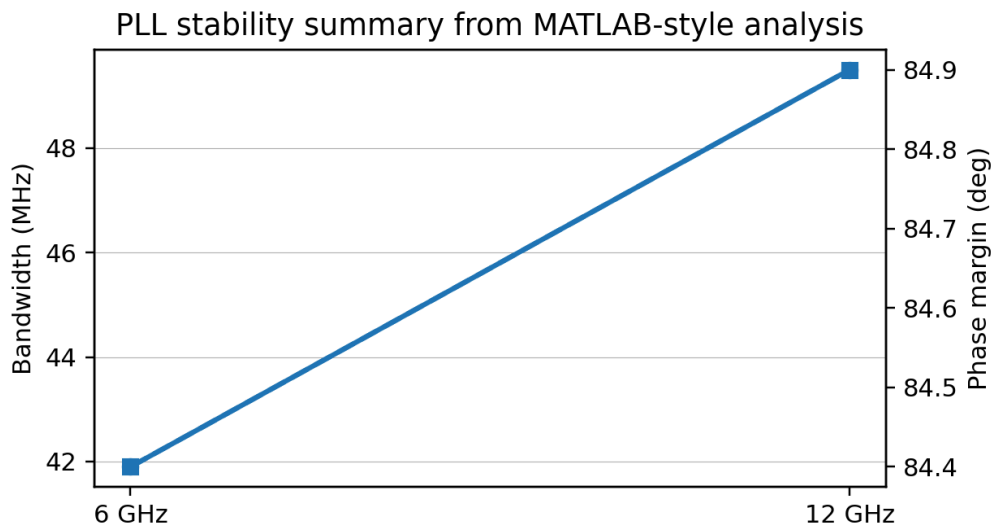
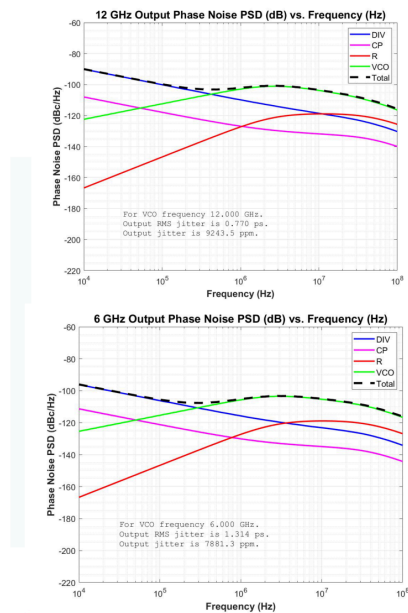


Figure 11. Summary of loop bandwidth and phase margin at the 6 GHz and 12 GHz corners.

The final performance evidence showed that the 12 GHz jitter was about 0.770 ps and the 6 GHz jitter was about 1.314 ps, both passing a 10,000 ppm specification. The 12 GHz total power was about 7.16

mW, and the 6 GHz total power was about 4.64 mW. These results demonstrated the engineering tradeoff between wide bandwidth, phase noise suppression, power, and control-voltage ripple.



Final Phase Noise Integration

- 12 GHz Performance: 0.770 ps (9,243 ppm) -> PASS
- 6 GHz Performance: 1.314 ps (7881 ppm) -> PASS
- Validation: 50+ MHz bandwidth successfully crushed VCO noise, 65-Ohm resistor kept thermal noise floored

Figure 12. Final phase-noise integration results showing jitter performance at 12 GHz and 6 GHz.

1.3 Smart Soil Moisture Sensor System

For sensor systems, I designed a smart soil moisture monitoring system using an ST0160 capacitive soil moisture sensor and an ESP32 microcontroller. The purpose of the system was to measure soil moisture for irrigation monitoring. This is a better portfolio example than a generic Wheatstone bridge because it shows a complete sensor system: sensing physics, signal conversion, ADC sampling, calibration, noise, and cross-sensitivity.



Figure 13. Smart soil moisture monitoring system block diagram showing sensor, analog voltage, ESP32 ADC, and wireless dashboard.

The sensor is capacitive. Its copper traces act as electrodes, and the surrounding soil acts as the dielectric material. The capacitance depends on $C = \epsilon A / d$. Since water has a much larger dielectric constant than dry soil or air, increasing soil water content increases the effective dielectric constant around the probe and changes the capacitance. The module converts this capacitance change into an analog voltage that the ESP32 ADC can read.

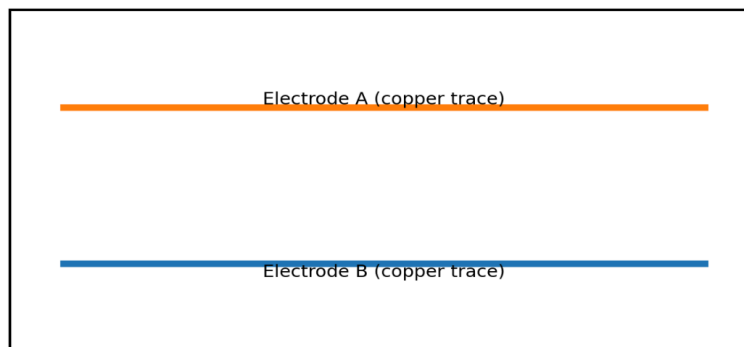


Figure 14. Physical model of the capacitive sensing region and explanation of dielectric-based sensing.

The sensor was selected because capacitive probes reduce corrosion compared with resistive probes. I classified the system as a capacitive, externally powered, analog-output, indirect field-monitoring sensor. The signal chain used a 3.3 V ESP32 environment, a 12-bit ADC, and a possible RC low-pass filter before the ADC. For a 3.3 V reference, one ADC count is approximately 0.805 mV.



Figure 15. Capacitive soil moisture sensor module used as the sensing element.

Calibration converted ADC counts into a practical moisture scale. Representative calibration data were: dry soil = 3000 counts, slightly moist = 2500 counts, moderate moisture = 2100 counts, wet = 1600 counts, and saturated soil = 1200 counts. Using the dry and saturated points, the sensitivity was calculated as $(1200 - 3000)/(100 - 0) = -18$ ADC counts per percent moisture. The negative sign means the ADC reading decreases as moisture increases for this module behavior.

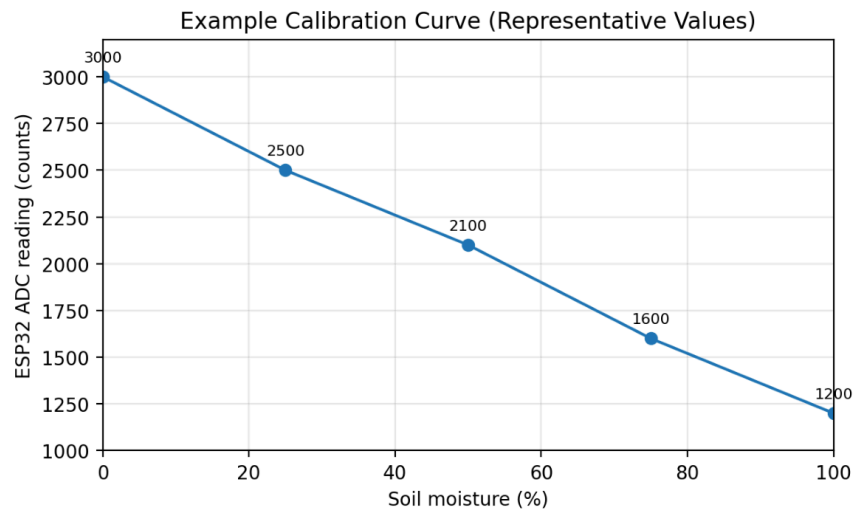


Figure 16. Calibration table and curve for the capacitive soil moisture sensor.

1.4 Communication Systems and Amplitude Modulation

In ECE 461/561, I simulated a double-sideband suppressed-carrier amplitude modulation (DSB-SC AM) system in Simulink. The model included a baseband source, carrier multiplication, additive Gaussian noise, coherent demodulation, and low-pass filtering. This example demonstrates my core knowledge in communication systems, frequency-domain analysis, noise, and filtering.

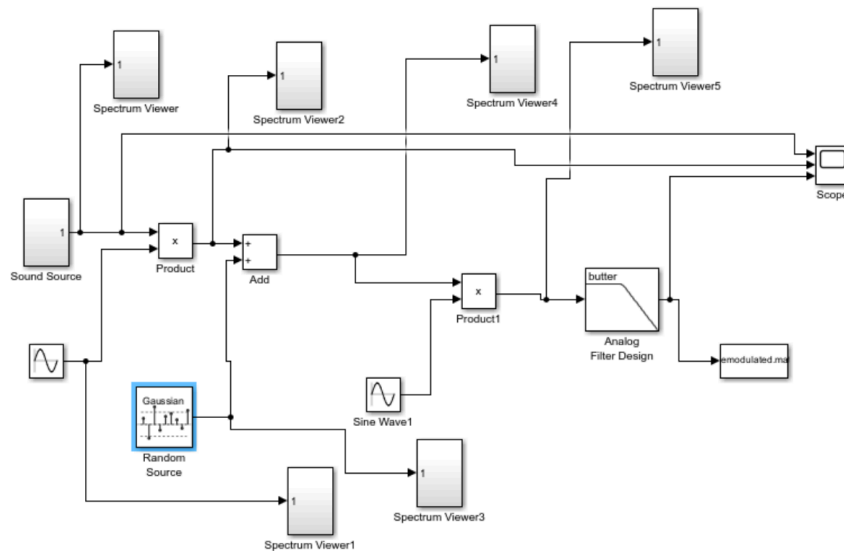


Figure 17. Simulink DSB-SC AM system model and theoretical derivation of coherent demodulation.

The transmitter multiplies the message signal $f(t)$ by a carrier $A \cos(\omega_c t)$, producing sidebands centered around the carrier frequency. In my simulation, the carrier frequency was 25 kHz. The spectrum showed the expected baseband content, carrier tones, and two shifted copies of the message spectrum at ± 25 kHz. This confirmed that the DSB-SC modulator was functioning correctly.

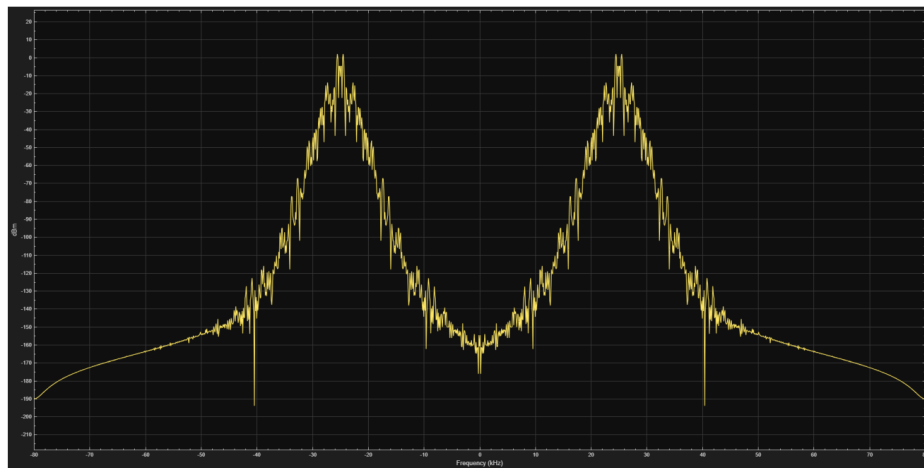


Figure 18. Spectrum evidence showing baseband and DSB-SC sidebands centered around the carrier.

At the receiver, coherent demodulation multiplied the received signal by a synchronized local carrier. The low-pass filter removed the high-frequency components around ± 50 kHz and recovered a scaled version of the original message. I tested filter cutoff choices and found that a 4 kHz low-pass edge produced the best sound quality because it passed the baseband audio while rejecting high-frequency noise and mixing products.

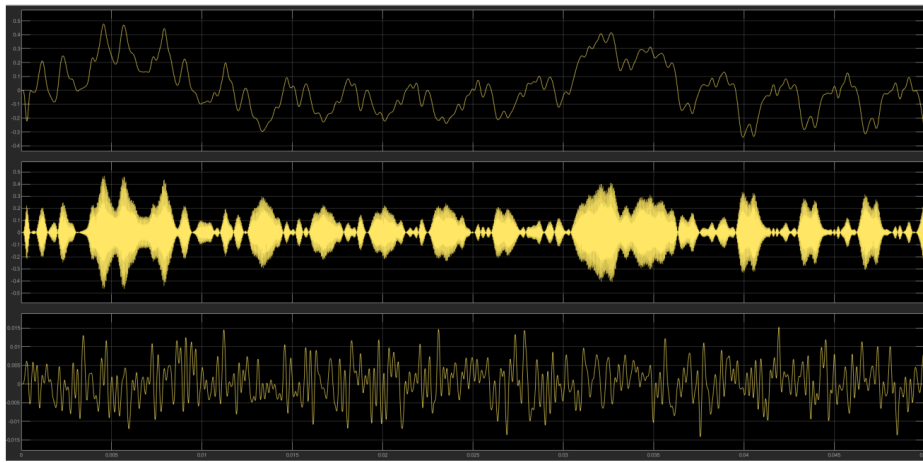


Figure 19. Recovered baseband waveform after coherent demodulation and low-pass filtering.

I also studied phase error in the local carrier. The theoretical output becomes $y(t) = (A/2) \cos(\theta) f(t)$, so phase mismatch reduces the recovered amplitude. Simulation results confirmed this: small phase error preserved the shape but reduced amplitude, $\pi/2$ phase error nearly removed the message, and π phase error inverted the waveform. This showed how synchronization accuracy affects communication-system performance.

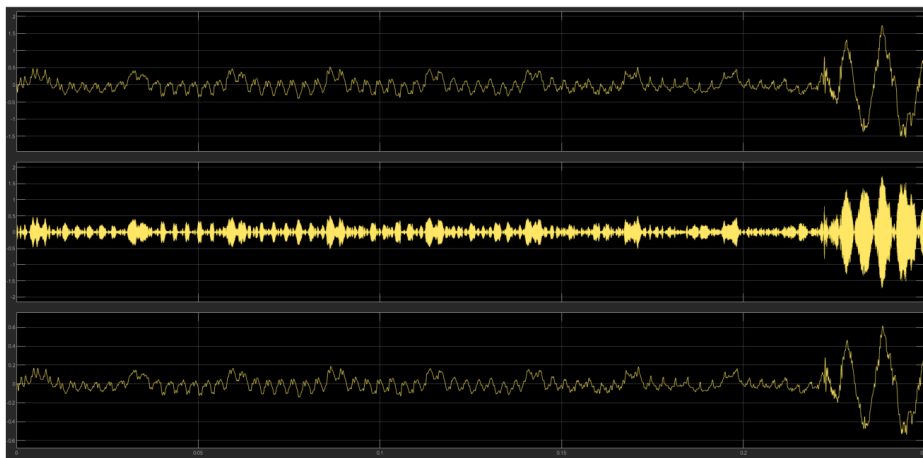


Figure 20. Phase-error demodulation result showing amplitude reduction caused by local-carrier phase mismatch.

In summary, Section 1 demonstrates that I can apply core ECE knowledge across devices, circuits, feedback systems, sensors, and communication systems. The evidence includes process simulation, transfer curves, PLL stability data, sensor calibration, and AM modulation/demodulation results.

Section 2: Integration of Core Knowledge with Other Fields

This section demonstrates how I integrated electrical engineering with physics, mathematics, statistics, and embedded systems. I focused on examples that have clear figures or measured/simulated evidence, because the portfolio should show how I used the knowledge rather than only define concepts.

2.1 Semiconductor Physics Integrated with CMOS Design

The CMOS process simulation integrated circuit design with semiconductor physics. The LOCOS result showed oxidation physics, where silicon dioxide grows vertically and laterally under the nitride mask. The final doping contour showed how implantation and diffusion created wells and source/drain junctions. These physical steps directly affected threshold voltage, leakage, capacitance, and device switching. By extracting NMOS and PMOS threshold voltages from transfer curves, I connected process variables to circuit operation. This is important because an inverter does not work correctly unless the fabrication process creates transistors with the intended polarity, threshold, and junction structure.

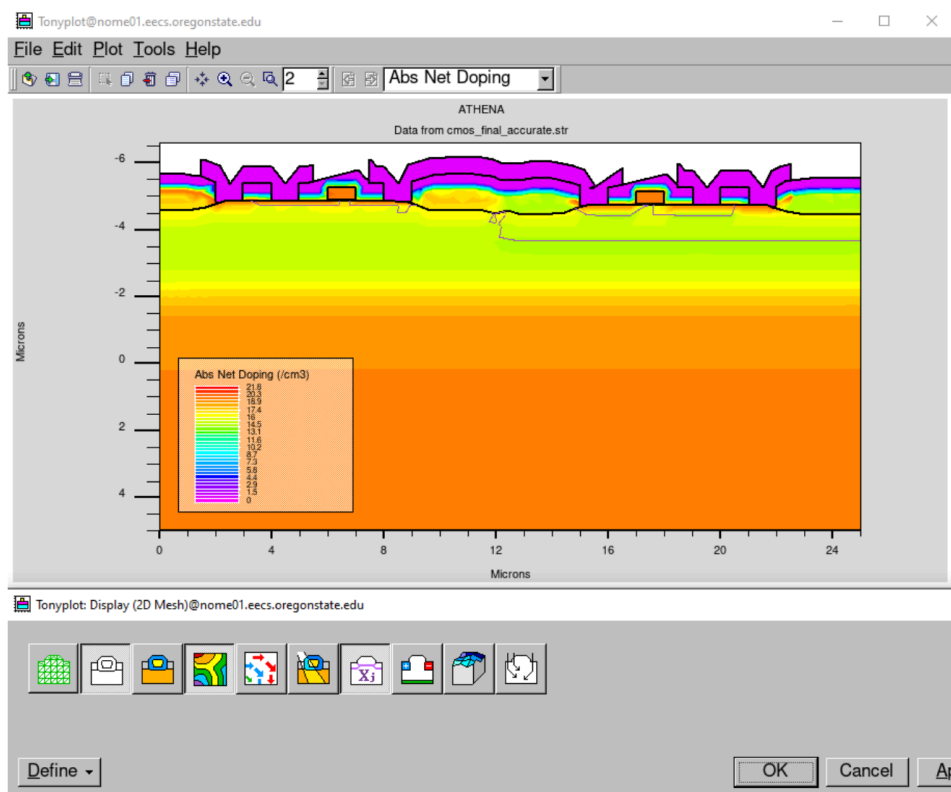


Figure 21. Integration of process physics and circuit design through final CMOS doping contours.

Extracted item	Engineering meaning	Portfolio evidence
Field oxide / LOCOS	Device isolation and leakage reduction	LOCOS and bird's-beak simulation
Gate oxide	Controls gate capacitance and electrostatic channel control	Gate oxide extraction in process simulation
Source/drain junctions	Affect parasitic resistance and short-channel behavior	Final structure and doping contour
NMOS/PMOS threshold voltage	Defines switching voltage and logic operation	Transfer curves and V_{th} extraction

2.2 Mathematics and MATLAB-Style Analysis Integrated with PLL Stability

The PLL example integrated circuit design with feedback control and mathematical modeling. The PLL loop can be represented by an open-loop gain $G(s)$ and closed-loop response $H(s) = G(s)/(1 + G(s))$, where s is the Laplace variable. The loop filter introduced a zero and a high-frequency pole to control bandwidth and phase margin. I used Bode-plot style verification to evaluate loop bandwidth and stability at both frequency corners. This directly addressed the faculty comment asking for MATLAB simulation evidence in Section 2.2.

Using the loop-filter values $R = 65$ ohms, $C1 = 1$ nF, and $C2 = 2$ pF, the design placed the zero near 2.4 MHz and the pole near 1.2 GHz. The 12 GHz case produced approximately 49.5 MHz bandwidth with 84.9 degrees phase margin, while the 6 GHz case produced approximately 41.9 MHz bandwidth with 84.4 degrees phase margin. These results show that math and simulation were used to choose a stable loop design rather than selecting values by trial and error.

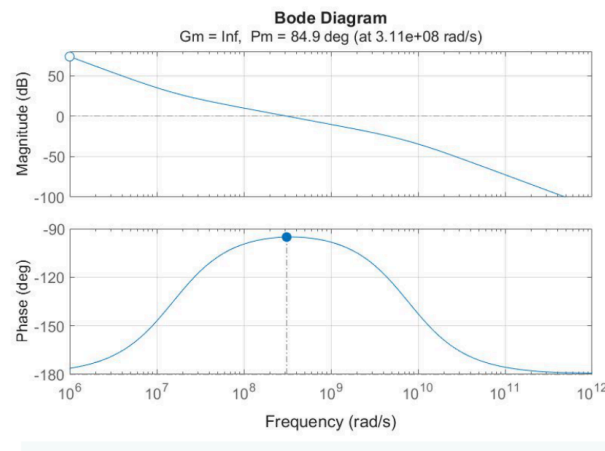


Figure 22. MATLAB-style Bode verification of PLL stability at the 12 GHz corner.

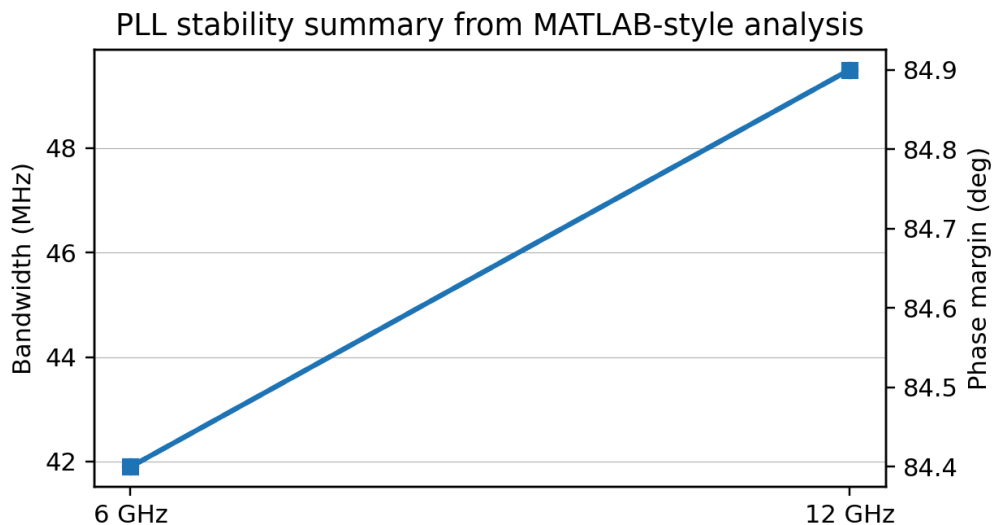


Figure 23. Numerical PLL stability summary comparing 6 GHz and 12 GHz operating corners.

2.3 Sensor Physics, Statistics, and Embedded Systems Integration

The smart soil moisture system integrated sensor physics, embedded systems, and statistics. The sensing physics came from dielectric permittivity: water changes the effective dielectric constant around the probe, which changes capacitance and the analog output voltage. The embedded-system part came from sampling the sensor with the ESP32 ADC and converting voltage into a digital count. The statistics part came from calibration, averaging, quantization noise, and uncertainty caused by temperature, soil type, density, salinity, and placement.

Using the representative calibration data, the system mapped ADC readings to a practical moisture scale. For example, $ADC_{dry} = 3000$, $ADC_{wet} = 1200$, and $ADC_{current} = 2100$ gave $Moisture = ((3000 - 2100)/(3000 - 1200)) \times 100 = 50\%$. With a sensitivity of -18 ADC counts per percent moisture, one ADC count corresponds to about 0.056% moisture. The ideal ADC quantization noise is much smaller than the practical accuracy limit, so the real design problem is not ADC resolution but calibration and cross-sensitivity.

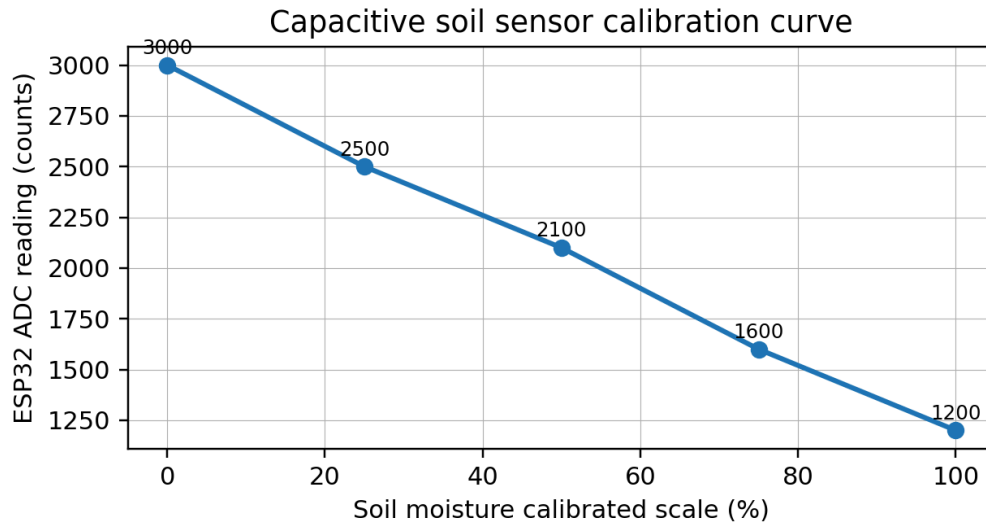


Figure 24. Sensor calibration curve generated from representative ADC values.

Cross-sensitive quantity	Effect	Mitigation
Temperature	Changes dielectric behavior and oscillator response	Add temperature sensor and compensate in software
Soil type	Clay, loam, and sand have different dielectric properties	Use soil-specific calibration curves
Soil density	Changes air gaps around sensor field lines	Install at consistent depth and avoid disturbing soil
Fertilizer / salinity	Changes electrical properties and apparent capacitance	Recalibrate after fertilizer changes
Sensor placement	Moisture varies with depth and root location	Use consistent placement or multiple sensors

2.4 Communication Theory Integrated with Simulation and Signal Processing

The amplitude modulation lab integrated mathematics, simulation, and signal processing. The DSB-SC signal was generated by multiplying the message by a carrier, which shifted the message spectrum to $\pm f_c$. In simulation, this produced sidebands at ± 25 kHz. During coherent demodulation, multiplying by the local carrier generated a desired baseband term and unwanted high-frequency terms at $\pm 2f_c$. The low-pass filter then recovered the message by passing 0-4 kHz and rejecting the components near 50 kHz.

This example also showed how noise and synchronization errors affect a real communication link. Additive Gaussian noise raised the noise floor, while local-carrier phase error scaled the recovered signal by $\cos(\theta)$. A frequency error produced a beating effect in the recovered audio. These results connect communications theory with practical simulation evidence.

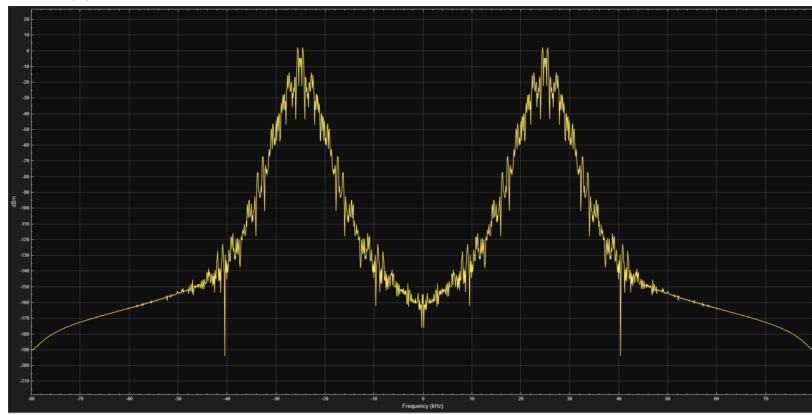


Figure 25. AM spectrum result showing sideband formation and frequency-domain verification.

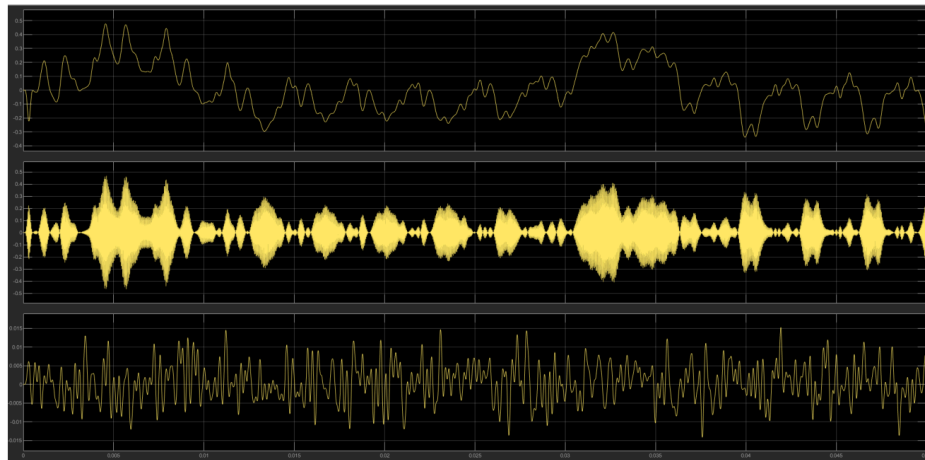


Figure 26. Demodulated baseband signal recovered by coherent mixing and low-pass filtering.

In summary, Section 2 demonstrates that I can combine ECE core knowledge with physics, mathematics, statistics, and signal processing. The evidence shows that I used modeling and simulation to analyze engineering behavior, not just describe concepts.

Section 3: Ethical Context of Engineering Practice

Engineering decisions affect safety, privacy, reliability, cost, and public trust. In my field, ethical engineering means designing circuits and systems that work as claimed, reporting limitations honestly, protecting user data, and avoiding misleading results. This is especially important in electronics because users often cannot directly see whether a system is safe or accurate; they must rely on the engineer's documentation and testing.

For the PLL example, ethical practice means clearly reporting tradeoffs such as high phase noise from a ring oscillator, reference spurs, static phase offset, and the increased power caused by the 2 mA charge pump. It would be misleading to report only that the jitter specification passed without also stating the cost in power and control-voltage ripple. For the soil moisture sensor, ethical practice means stating that the values are representative calibration data unless verified by experimental measurement. If the sensor is used for irrigation, inaccurate readings could waste water or damage plants, so calibration and cross-sensitivity must be disclosed. For the AM communication lab, ethical practice means correctly reporting the effect of noise, phase error, and frequency error rather than hiding degraded performance.

The IEEE Code of Ethics emphasizes honesty, realistic claims, safety, and avoiding harm. I applied these ideas by labeling representative values, explaining limitations, and showing actual figures and simulation outputs as evidence. In future engineering work, I will continue to document assumptions, test limits, cite sources, and communicate uncertainty clearly so that design decisions are transparent and responsible.



Figure 27. CITI Certificate.

References

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- [2] B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill, 2017.
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